MODULE II

APPLICATIONS OF BIOMOLECULES

Contents: Carbohydrates in cellulose-based water filters production, PHA and PLA in bioplastics production, Nucleic acids in vaccines and diagnosis, Proteins in food production, lipids in biodiesel and detergents production, Enzymes in biosensors fabrication, food processing, detergent formulation and textile processing.

CARBOHYDRATES:

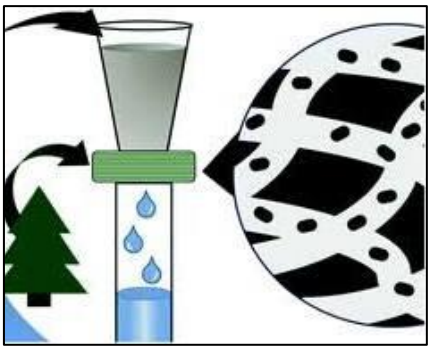
- Carbohydrates are molecules made up of carbon, hydrogen, and oxygen atoms.
- They are a major source of energy for the body.

CELULLOSE:

- Cellulose is a type of complex carbohydrate made of many glucose units.
- It's found in large amounts in nature, is renewable, and can break down naturally.
- Cellulose doesn't dissolve in water or common organic solvents because of how its glucose units are bonded together.
- Taking benefit of these advantages of cellulose, we have a best application of cellulose that is Cellulose based water filters.

CELLULOSE-BASED WATER FILTERS:

- *Cellulose filter paper is a type of filter paper made from cellulose fibers, which are derived from plants.*
- Interest in biobased filters for water purification is growing due to their potential for affordability, light weight, and biodegradability.
- Research focuses on creating biobased membranes from cellulose nanofibrils (CNFs) for micro and ultrafiltration.



Properties Of Cellulose Based Water Filter Papers:

- **High porosity:** Cellulose fibers have a naturally high porosity, which allows water to flow through the filter easily. This makes them ideal for applications where fast flow rates are desired.
- **Biodegradable:** Cellulose is a natural polymer that is biodegradable, making cellulose-based water filter paper an environmentally friendly option.
- **Low cost:** Cellulose is a relatively abundant and inexpensive material, making cellulose-based water filter paper an affordable option for low-resource settings.
- **Limited size exclusion:** While cellulose-based filter paper can remove particles due to size exclusion, the pores are often large enough to allow bacteria to pass through.
- Cellulose-based water filter paper can be modified to improve its performance.
  For example, it can be impregnated with silver nanoparticles to kill bacteria or coated with a charged polymer to remove contaminants through electrostatic interactions.

Advantages of Cellulose Based Water Filter Papers:

- **Affordability:** Cellulose is a naturally abundant and inexpensive material, making these filters a cost-effective solution for clean water access.
- **Biodegradability:** Cellulose decomposes naturally, minimizing waste disposal concerns compared to synthetic filter materials.
- **Availability:** Cellulose is a widely available material, making these filters obtainable in remote areas where access to specialized filtering equipment might be limited.
- **Reasonable Flow Rate:** The high porosity of cellulose allows for a decent flow rate during filtration, making it practical for everyday use.
- **Large Particle Removal:** Cellulose filter paper can effectively trap larger particles like sediments, debris, and even some parasites from the water, improving clarity and reducing turbidity.

POLY LACTIC ACID [PLA] AND POLY HYDROXY ALKANOATES [PHA]:

- **PLA (Poly Lactic Acid)** and **PHA (Poly Hydroxyalkanoates)** are bioplastics that serve as eco-friendly alternatives to traditional, non-biodegradable plastics.
- ***Bioplastics are one type of plastic which can be generated from natural resources such as starches and vegetable oils.***
- Bioplastics are derived from natural resources like starches and vegetable oils, and they are classified as bio-based and/or biodegradable.
- Not all bio-based plastics are biodegradable, and not all biodegradable plastics are bio-based.
- Bioplastics are considered bio-based when the carbon building blocks originate from natural sources, regardless of their biodegradability.
- Bioplastics are biodegradable if they can be broken down by environmental conditions and microbes, serving as a food source for them.
- Bioplastics are deemed compostable if they can be fully assimilated by microbes in a compost environment within 180 days.

COMPARISON BETWEEN PLA (POLY LACTIC ACID) AND PHA (POLY HYDROXY ALKANOATES):

PROPERTIES	PLA	PHA
SOURCE MATERIAL	Made from renewable resources like corn starch, sugarcane, or cassava.	Produced by microorganisms feeding on sugars or organic matter.
BIODEGRADABILITY	Biodegradable under industrial composting conditions (high temperatures). May take a long time to break down in a natural environment.	Generally considered more biodegradable. Can break down in various conditions, including soil and marine environments.
MECHANICAL PROPERTIES	Stiffer and more brittle than PHA. Similar properties to PET (polyethylene terephthalate) plastic used in bottles.	More flexible and ductile than PLA. Offers a wider range of possible properties depending on the type of PHA.
COST	Currently less expensive to produce than PHA due to established production methods.	More expensive due to relatively new production processes.
PRODUCTION	Requires high temperatures, pressures, and catalysts during the polymerization process. Can be energy-intensive.	Produced through fermentation by microorganisms. Generally, less energy-intensive than PLA production.

NUCLEIC ACIDS:

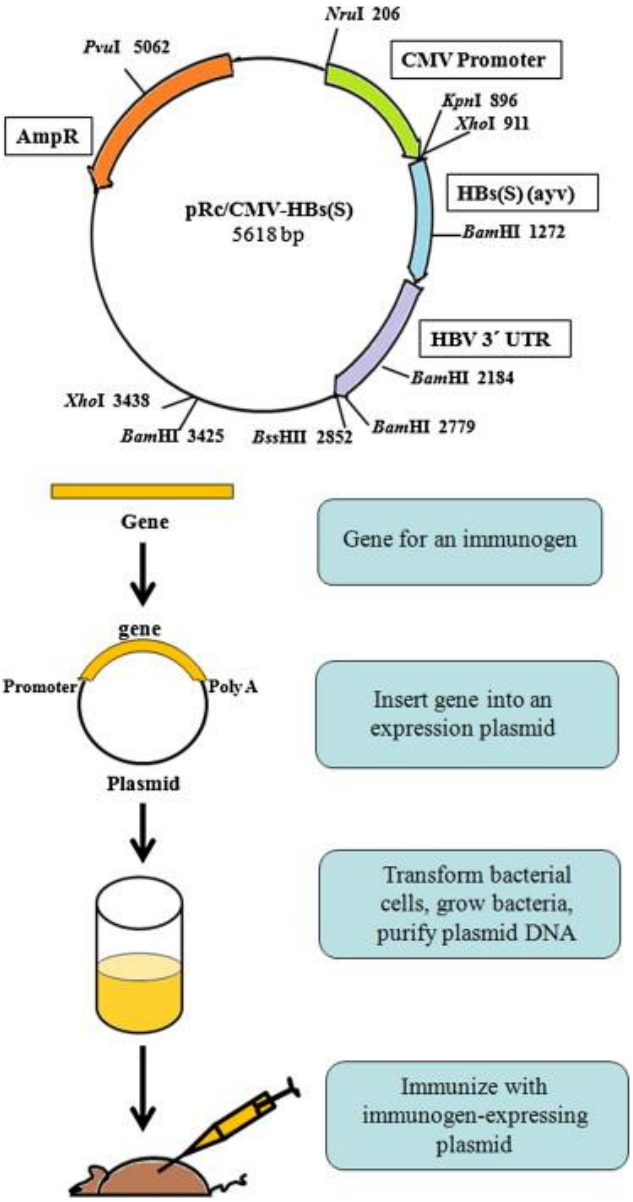
- Nucleic acids are macromolecules essential for life.
- They are made up of nucleotides, consisting of a sugar, phosphate group, and nitrogenous base.
- The two main types of nucleic acids are DNA and RNA.
- DNA carries genetic information, while RNA helps in protein synthesis.

DNA VACCINE FOR RABIES:

- DNA vaccine for rabies uses a plasmid encoding the rabies virus glycoprotein gene.
- It induces long-lasting protective immunity against rabies virus in dogs.
- The vaccine has shown safety and efficacy in preventing rabies.
- It offers advantages like rapid development and long-lasting immunity.
- Research continues to enhance its effectiveness for human use.

STEPS INVOLVED IN DNA VACCINE DEVELOPMENT FOR RABIES:

1. **Designing the Plasmid:** The plasmid is engineered to contain the gene for the rabies virus glycoprotein. This gene is selected because it is a key protein that stimulates the immune system to produce antibodies against the rabies virus.
2. **Inserting the Gene:** The gene for the rabies virus glycoprotein is inserted into the plasmid using molecular biology techniques. This ensures that the plasmid will carry the gene into the host's cells.
3. **Vaccine Preparation:** The plasmid containing the gene is purified and prepared for vaccination. It is often mixed with other substances to enhance its stability and effectiveness.
4. **Administration:** The vaccine is administered to the host, typically through injection. The plasmid enters the host's cells, where the gene for the rabies virus glycoprotein is expressed.
5. **Immune Response:** The host's immune system recognizes the glycoprotein as a foreign substance and mounts an immune response. This includes the production of antibodies specific to the glycoprotein.
6. **Protection:** The antibodies produced by the immune system are capable of recognizing and neutralizing the rabies virus. This provides the host with protection against rabies if they are exposed to the virus in the future.
7. **Monitoring:** The immune response and any side effects of the vaccine are monitored to ensure its safety and effectiveness.



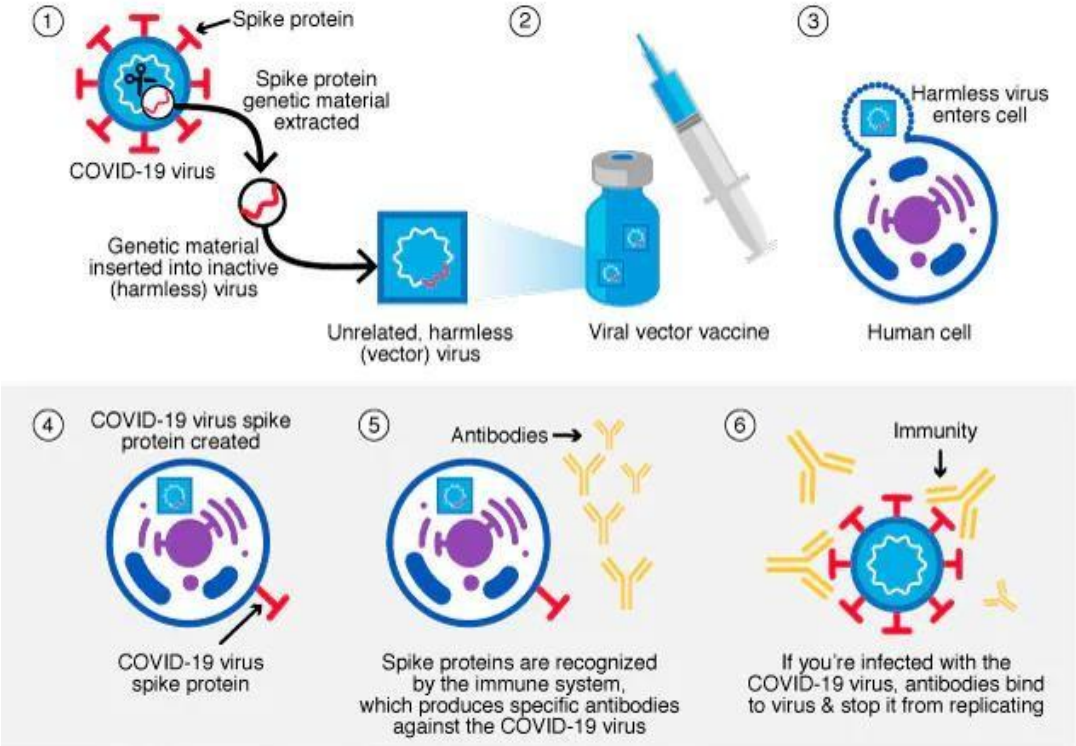
RNA VACCINE FOR COVID:

- COVID-19 is caused by the SARS-CoV-2 virus and mRNA technology is used in vaccines to generate an immune response.
- mRNA instructs cells to produce a protein that triggers antibody production, providing protection against COVID-19.
- Pfizer/BioNTech and Moderna vaccines use mRNA technology, part of the COVAX global vaccine initiative.
- mRNA vaccines do not alter DNA, as the mRNA is broken down after protein production.
- The immune system recognizes the protein produced and builds a response to fight the virus.

STEPS INVOLVED IN RNA VACCINE DEVELOPMENT FOR COVID-19:

1. **mRNA Design:** Scientists design the mRNA sequence to encode a specific viral protein, in this case, the spike protein of SARS-CoV-2.
2. **mRNA Production:** The mRNA is synthesized in the laboratory using a template that matches the viral protein sequence.
3. **Lipid Nanoparticle Encapsulation:** The mRNA is encapsulated in lipid nanoparticles to protect it and help it enter cells.

4. **Vaccination:** The mRNA vaccine is administered through injection, typically into the muscle.
5. **Cellular Uptake:** Cells at the injection site take up the lipid nanoparticles containing the mRNA.
6. **Protein Production:** The mRNA instructs the cells to produce the spike protein of the virus.
7. **Immune Response:** The immune system recognizes the spike protein as foreign and mounts an immune response, including the production of antibodies.
8. **Memory Cells:** Some of the immune cells generated "remember" the spike protein, providing immunity in case of future exposure to the virus.



PROTIENS:

- Proteins are essential biomolecules present in all living organisms, vital for numerous biological functions.
- They consist of long chains of amino acids, with each protein having a specific sequence that determines its function.
- Proteins play critical roles in catalyzing chemical reactions, providing structural support, and transporting molecules within cells.

PROTEIN AS FOOD:

- Protein is a crucial component of a balanced diet, with the average person requiring about 7 grams per day for every 20 pounds of body weight.
- While many foods contain protein, it's important to consider the overall nutritional value of the food.
- Animal-based foods like meat, poultry, fish, eggs, and dairy are good sources of complete protein, containing all essential amino acids.
- In contrast, plant-based foods like fruits, vegetables, grains, nuts, and seeds may lack certain essential amino acids, making it important to eat a variety of plant-based proteins to ensure adequate intake.

WHEY PROTEIN:

- **Definition & Composition:** Whey protein is a mixture of proteins derived from whey, a by-product of cheese production. It contains α-lactalbumin, β-lactoglobulin, serum albumin, and immunoglobulins. Glycomacropeptide, though present, is not a protein.
- **Production Process:** Whey is the liquid left after **milk coagulation** during cheese production. It contains **water-soluble proteins, lactose, and some lipids**. The pH is reduced to **4.6**, separating whey from curd. Processing involves drying or removing **lactose, lipids, and non-protein materials** to concentrate protein content.
- **Uses & Benefits:** Whey protein is widely used as a **dietary supplement** for **muscle growth and recovery**. While consuming it before exercise does not directly improve performance, it enhances **protein synthesis and recovery** after workouts by increasing **free amino acids** in the body.
- Whey protein can be processed to increase its protein content by removing lactose, lipids, and other non-protein materials.

MEAT ANALOGUES:

- Meat analogues are gaining popularity among consumers seeking **healthy, ethical, cost-effective, and environmentally friendly** alternatives to conventional meat. These products mimic meat in **appearance, texture, and sensory attributes**, providing a similar **eating experience and nutritional value**.
- A key technology in producing meat analogues is **high-moisture extrusion cooking**, which helps create **muscle meat-like texture** using **plant or animal proteins**. The final product looks and feels like **cooked meat**, with a **high protein content** comparable to conventional meat.
- The shift towards meat analogues is driven by factors such as **low-fat and low-calorie food preferences, flexitarian diets, concerns over animal diseases, depletion of natural resources, and the need to reduce greenhouse gas emissions**.
- Currently available meat analogues are mostly **plant-based**, with ingredients like **soy proteins, mycoprotein, and soy leghemoglobin** enhancing texture and taste. Key aspects of these products include **ingredients and structure formation, flavour, taste, nutritional value, post-extrusion processing, packaging, shelf life, consumer benefits, and environmental impact**.

Meat analogues provide a **sustainable and nutritious** alternative, addressing consumer demands while reducing the environmental footprint of food production.

PLANT BASED PROTIENS:

- **Definition & Sources:** Plant proteins come from **pulses, tofu, soy, tempeh, seitan, nuts, seeds, grains, and peas**. Pulses include **chickpeas, lentils, black beans, kidney beans, adzuki beans, and split peas**.
- **Nutritional Benefits:** They are excellent sources of **protein, fibre, vitamins, and minerals**. Consuming **pulses, peas, and nuts** can help increase **fibre intake**, which is often insufficient in many diets.
- **Rising Consumer Demand:** The demand for plant-based proteins is growing due to several factors:

    1. **Health benefits** of plant-based diets.

2. **Concerns about the negative effects** of animal protein, such as high saturated fat.
3. **Environmental sustainability** in food production.
4. **Ethical concerns** about animal treatment.
5. **General perception of protein as a beneficial nutrient** for health and physical function.

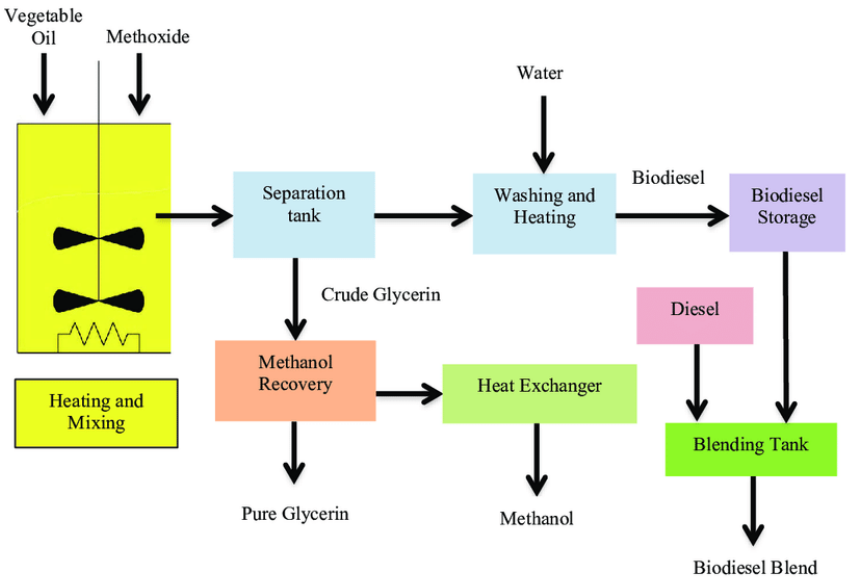
- **Nutritional Considerations:** While **plant proteins offer many benefits**, they may have **some limitations** in nutritional quality compared to animal proteins.
- Consumer demand for plant protein-based products is high and expected to grow significantly in the next decade.
- While diets high in plant-based protein have health benefits, plant proteins may be inferior to animal proteins in some aspects of nutritional quality.

LIPIDS:

- Lipids are a diverse group of molecules including fats, waxes, sterols, and phospholipids.
- They serve functions such as energy storage, cell membrane structure, and signaling.
- Lipids are hydrophobic or amphiphilic, allowing them to form structures in water.
- They originate from biochemical subunits ketoacyl and isoprene groups.
- Lipids have applications in industries like cosmetics, food, and nanotechnology.

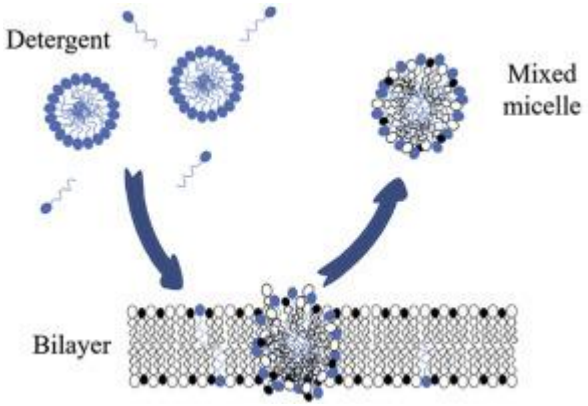
BIODIESEL:

- Lipids obtained from food waste can serve as a valuable feedstock for biodiesel production. This process involves transesterification, where the lipids are reacted with an alcohol (such as methanol or ethanol) in the presence of a catalyst to produce biodiesel and glycerol as a byproduct.
- **Chemical Catalyst:** Using potassium hydroxide in the transesterification process can achieve a 100% conversion of lipids to biodiesel in a relatively short time, typically around 2 hours.
- **Biocatalyst:** Biocatalysts like Novozyme-435, an enzyme, can also be used for transesterification. This method is more environmentally friendly, it may take longer (up to 24 hours) and yield slightly lower conversion rates, around 90%.
- Biodiesel, a renewable and biodegradable fuel, is produced from animal fats or vegetable oils, including soybean oil, vegetable oil, and animal fat (tallow). It can be used in standard diesel engines without modification. Biodiesel blends can also be used as heating oil or blended with gasoline in any proportion.



DETERGENTS:

1. Detergents are a type of surfactant, or surface-active agent, that are widely used for cleaning and removing dirt and grease.
2. Detergents are like tiny cleaning molecules that can disrupt the structure of cell membranes, which are like protective barriers around cells.
3. They have a part that loves water (hydrophilic head) and a part that hates water (hydrophobic tail).
4. The hydrophobic tail of the detergent can sneak into the cell membrane's oily interior, causing it to break apart.
5. This breaking apart of the membrane helps detergent molecules to grab onto and dissolve membrane proteins and other stuff from the membrane.
6. In water, detergent molecules can also form tiny clusters called micelles, where the hydrophobic tails hide in the middle and the hydrophilic heads stick out, making them soluble and able to carry away the membrane components.
7. Scientists use this property of detergents to study and extract proteins from cell membranes for research and other purposes.
8. Detergents and lipids both have a similar structure, with a polar head and a nonpolar tail. However, they differ in the shape of their monomers, the way they form groups in a solution, and the amount needed for these groups to form.



ENZYMES:

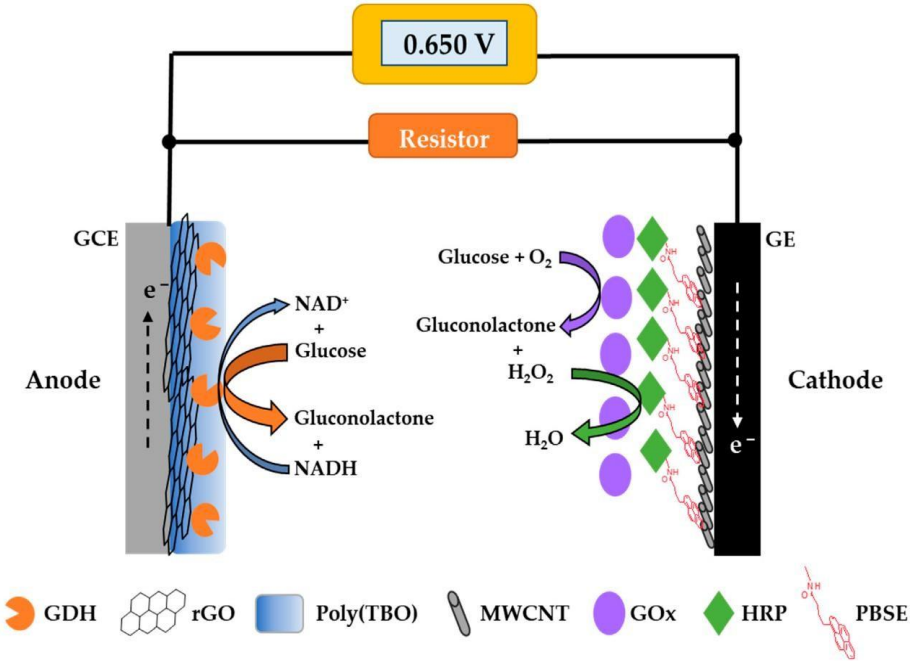
- Enzymes are proteins that accelerate chemical reactions in living organisms.
- They lower the activation energy required for reactions to proceed.
- Enzymes are essential for metabolism, building some substances and breaking down others.
- Living organisms naturally produce enzymes.
- There are six main types of enzymes: hydrolases, oxidoreductases, lyases, transferases, ligases, and isomerases.
- Enzyme-based biosensors offer fast, precise, and continuous monitoring of analytes due to their high specificity, simplicity, and scalability.

ENZYMES IN BIOSENSORS:

- Biosensors are utilized in disease monitoring, drug discovery, and detecting pollutants and disease-causing microorganisms.
- They are also used to detect markers that indicate a disease in bodily fluids like blood, urine, saliva, and sweat.
- Different types of biosensors include enzyme-based biosensors, immunosensors, DNA biosensors, thermal biosensors, and piezoelectric biosensors.
- Enzymes play a crucial role in biosensors, with glucose oxidase being a common enzyme used in amperometric glucose biosensors.

GLUCOSE OXIDASE IN BIOSENSORS:

- Glucose oxidase (Gox) is a commonly used enzyme in glucose biosensors due to its stability and cost-effectiveness.
- Gox facilitates the redox reaction and transfers electrons from its active sites to the electrode for analyzing glucose levels in blood samples.
- Amperometric glucose biosensors are created by immobilizing Gox, which can reduce oxygen to hydrogen peroxide and convert glucose to d-glucono-1,5-lactone.
- Glucose levels can be quantified by detecting the hydrogen peroxide produced or the oxygen consumed in this process.



ENZYMES IN FOOD PROCESSING:

Enzymes are like tiny helpers in food processing. They speed up important reactions without getting used up themselves. This makes food processing faster, improves quality, and can even lead to new food products.

Benefits of Using Enzymes:

- **Enhanced Efficiency:** Enzymes can significantly speed up reactions, leading to faster processing times, increased production output, and reduced energy consumption.
- **Improved Quality:** Enzymes can help control specific reactions, leading to more consistent and desirable textures, flavors, and colors in food products.
- **Nutritional Enhancement:** Enzymes can break down complex molecules into simpler, more bioavailable forms, improving the nutritional value of food.
- **Creation of New Products:** Enzymes can be used to develop entirely new food products or modify existing ones to meet specific dietary needs or preferences.
- **Reduced Waste:** Enzymes can help extract valuable components from raw materials, minimizing waste generation.

Common Applications of Enzymes in Food Processing:

- **Baking Industry:** Amylases break down starches into sugars, allowing for dough rise and fermentation in bread production.

- **Dairy Industry:** Rennet (chymosin) separates milk curds from whey in cheesemaking. Lactase breaks down lactose sugar in milk for lactose-intolerant individuals.
- **Beverage Industry:** Pectinases clarify fruit juices and improve filtration efficiency.
- **Fruit and Vegetable Processing:** Pectinases and cellulases help extract juices, improve texture, and reduce viscosity.
- **Fats and Oils:** Lipases break down fats and oils, modifying texture and flavour in products like margarine.

ENZYMES IN TEXTILE PROCESSING:

Enzymes are revolutionizing the textile industry by offering a more sustainable and efficient alternative to traditional chemical processes. Here's a breakdown of their key roles:

Benefits of Enzymes:

- **Eco-friendly alternative:** Enzymes are replacing harsh chemicals in many textile processes, reducing pollution and making the industry more sustainable.
- **Specificity:** Enzymes act on specific targets like starch or hydrogen peroxide, without harming the fibres themselves.
- **Mild conditions:** Enzymes work effectively at lower temperatures compared to traditional methods, reducing energy consumption.

Types of Enzymes Used:

Different enzymes are used for various stages of textile processing, including:

- **Desizing:** Amylase enzymes break down starch-based sizing applied to yarns during weaving.
- **Scouring:** Cellulase and pectinase enzymes remove natural impurities like waxes and pectins from cotton fibres.
- **Bleaching:** Laccase enzymes help bleach textiles and degrade excess hydrogen peroxide used in the process.
- **Denim finishing:** Cellulase enzymes create a worn-out, distressed look on denim fabrics.
- **Wastewater treatment:** Lactase enzymes can decolorize textile dyes in wastewater, making it more eco-friendly.

ENZYMES IN DETERGENT FORMULATION:

Enzymes play a crucial role in modern detergent formulations, offering several advantages over traditional cleaning methods. Here's a breakdown:

Types of Enzymes in Detergents:

- **Proteases:** These enzymes break down proteins, tackling stains like blood, grass, sweat, and food residues.
- **Lipases:** They specialize in breaking down fats and oils, effectively removing greasy stains and food residue.
- **Amylases:** These enzymes target starch-based stains like sauces, gravies, and cereal residue.
- **Mannanases:** These enzymes break down hemicelluloses, a component of certain food stains and plant-based materials.

Benefits of Using Enzymes in Detergents:

- **Enhanced Stain Removal:** Enzymes specifically target and break down various types of stains, leading to more effective cleaning.
- **Lower Washing Temperatures:** Enzymes work effectively at lower temperatures, saving energy and reducing wear and tear on fabrics.
- **Environmental Advantages:** Using enzymes can reduce the need for harsh chemicals in detergents, making them more environmentally friendly.
- **Improved Fabric Care:** Cellulases can help maintain the softness and texture of fabrics.

How Enzymes Work in Detergents:

1. **Stain Breakdown:** Enzymes act as biological catalysts, breaking down complex stains into smaller, water-soluble components.
2. **Surfactant Action:** Detergents contain surfactants that lift and suspend the broken-down stain particles, allowing them to be rinsed away.
3. **Effective at Lower Temperatures:** Enzymes function optimally at lower temperatures compared to traditional cleaning methods, leading to energy savings.

QUESTION BANK:

1. What is Cellulose-Based water filters? Explain properties of cellulose based water filters.
2. Explain advantages of Cellulose-Based water filters.
3. What are bioplastics? Distinguish between PLA and PHA bioplastics.
4. Explain DNA vaccines for Rabies?
5. Explain RNA vaccines for Covid-19?
6. Explain Whey Protein.
7. Explain Meat analogues and Plant based proteins.
8. Explain the production of biodiesel from lipids.
9. Describe Detergents.
10. What are Enzymes? Explain the role of enzymes in biosensors.
11. Explain Glucose Oxidase in Biosensors.
12. Explain the role of enzymes in food processing.
13. Explain the role of enzymes in textile processing.
14. Explain the role of enzymes in detergent formulation.